Friction: before and during sliding

LEARNING TARGET I can distinguish adjustable static friction from kinetic friction.

A · KEY VOCABULARY

Static friction	Before sliding. Adjusts to match the pull, up to a limit.
Maximum static	The greatest static friction possible. Pull past it and sliding starts.
Kinetic friction	During sliding. Use the supplied value.
Coefficient (μ)	When given: kinetic friction = μ × normal force.

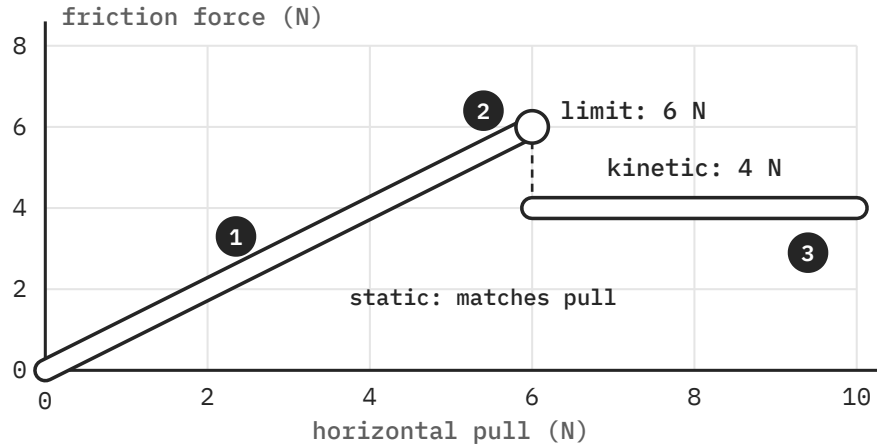
B · KEY EQUATIONS + UNITS

$$\text{at rest: } f_s = \text{pull } (\le \text{limit})$$
$$\text{sliding: } f_k = \mu \times F_N$$

friction	newtons (N)
μ	no unit

C · COLOR IT · BEFORE AND DURING SLIDING

Example crate: limit 6 N, kinetic 4 N.



- ☐ **1 · ORANGE** Static friction: matches pull
- ☐ **2 · ORANGE** The limit (dashed)
- ☐ **3 · ORANGE** Kinetic friction

D · WORKED EXAMPLE

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration.

GIVEN	EQUATION	WORK	ANSWER
m = 2 kg pull 10 N right f_k 4 N left	$a = F_{net} \div m$	$F_{net} = 10 - 4 = 6$ N right $a = 6 \div 2$	**3 m/s² right**

E · THE 2026-XJ THREAD

DAY 17 OF 22 · NEW TOOL: FRICTION

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else